



CONTENT



RESTORATION



AUTHORSHIP



FAMILIARITY



TEST

 **Name: Roy Batty**

Student Number: N6MAA10816

**[01] I've seen things you people wouldn't believe. [02] Attack ships on fire
off the shoulder of Orion. [03] I watched C-beams glitter in the dark near the
Tannhäuser Gate. [04] All those moments will be lost in time, like tears in rain.**

● ● ●
Name: Roy Batty

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Sentence Intruders & Synonym Replacements

Extra sentences have been added. Circle the added sentence numbers.

Five words have been replaced. Find the replacements and provide the originals.

Incorrect sentence numbers (−1 point). Incorrect word choices (0 points).

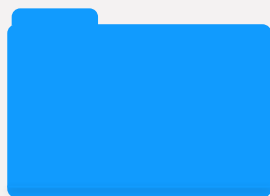
[01] I've seen things you people wouldn't believe. [02] These experiences include witnessing combat ships aflame beyond the reach of conventional rescue operations. [03] Attack ships on fire off the edge of Orion. [04] I observed C-shafts glitter in the dark near the Tannhäuser Gate. [05] All those moments will be lost in time, like weeping in downpour. [06] These fleeting experiences underscore the fragile impermanence of memory amid the relentless passage of time.

part of speech	noun	noun	verb	noun	noun
replacements					
originals	b	t	w	s	r



The Problem

LLM-influenced
Writing may
not reflect
learning
outcomes



Potential Solutions

Artifact-based
approaches &
process-based
approaches



Research Methods & Procedures

within subjects
design;
handwritten text
and unseen LLM
text baselines



Results & Discussion

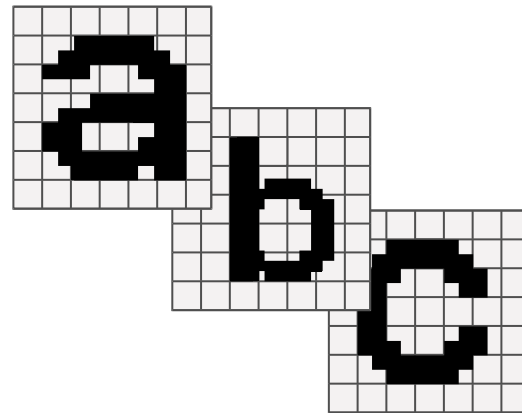
non-overlapping
performance
distribution,
large effect size

The Problem

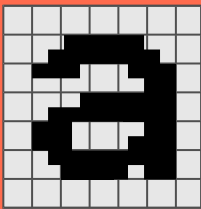
Authorship

When LLMs (e.g., ChatGPT) can generate academic text, writing no longer reliably indicates authorship or learning.

- Students cannot clearly judge their own contribution to LLM-assisted work.
- Institutional responses (bans, detection, disclosure) are unreliable and hard to enforce.

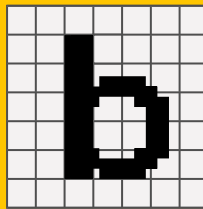


Artifact-based Detection



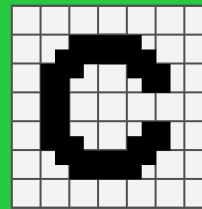
Classifier-based Detection

- + Fast and scalable
- + Trained on human vs. AI text data
- Probabilistic, not proof of authorship
- Sensitive to context (domain, population)



Stylometry-based Detection

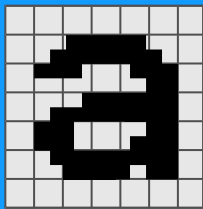
- + Interpretable linguistic features
- + Can capture consistent writing patterns
- Assumes stable individual style
- Weakened by editing or AI assistance



Transformer-based Detection

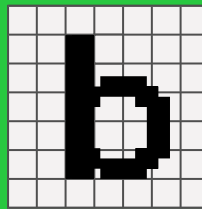
- + Leverage contextual and semantic signals
- + Strong performance in controlled settings
- Dependent on training data/conditions
- Limited generalizability across contexts

Evading Artifact Detection



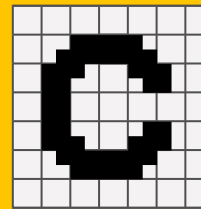
Rewriting attacks

- Paraphrasing disrupts detection
- Accessible paraphrase tools
- Shifts lexical/syntactic patterns



Editing attacks

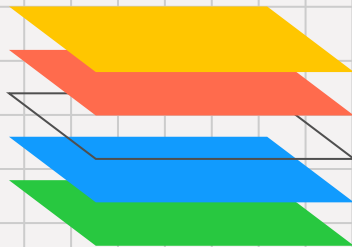
- Small surface changes undermine model predictions
- May not affect readability
- Low-effort, high-impact evasion



Prompt attacks

- Prompt variation produces alternative outputs
- Iterative refinement reduces detectability

Authorship as a Process



Memory

Authorship is **not inherent** in the text, but reflects a writer's **cognitive relationship** with the text, as shaped by **decisions** made during **planning, drafting, and revision**.

Planning

- Organizing ideas
- Setting goals for expression
- Encoding intended meaning & rhetorical structure

Drafting

- Transforming ideas into text forms
- Actively selecting words & structures
- Encoding formulation decisions

Revision

- Evaluating and refining lexical and structural choices
- Aligning text with intended meaning and goals
- Re-evaluating prior decisions

Checking the Writing Process

Words

- Writers encode specific word choices during planning, drafting, and revision processes
- Writers should notice when their carefully selected words are replaced with synonyms
- CRAFT asks students to identify 5 original and 5 replacement words (10 points)
- Hints provided (first letter, part of speech); no penalty for incorrect responses

Structure

- Writers encode discourse and organizational decisions during planning, drafting, and revision processes
- Writers should notice when sentences do not fit their structure
- CRAFT asks students to identify four (4) LLM-generated intruder sentences (4 points)
- Incorrect selections are penalized (-1 point) to discourage guessing

Research Methods & Procedures

Human Condition

- Participants compose sixteen (16) sentences by hand on paper, devices are prohibited
- Participants transcribe their text and then a Python script creates a personalized CRAFT test for each participant
- About 15 minutes after writing, participants restore their lexical and structural decisions
- Participants have ten (10) minutes to complete the test

LLM Condition

- An LLM generates sixteen (16) sentences on the same topic as the handwritten text
- Generated text is used to create a personalized CRAFT test using the same Python script
- With no knowledge of the text, participants try to identify lexical and structural changes
- Participants have ten (10) minutes to complete the test

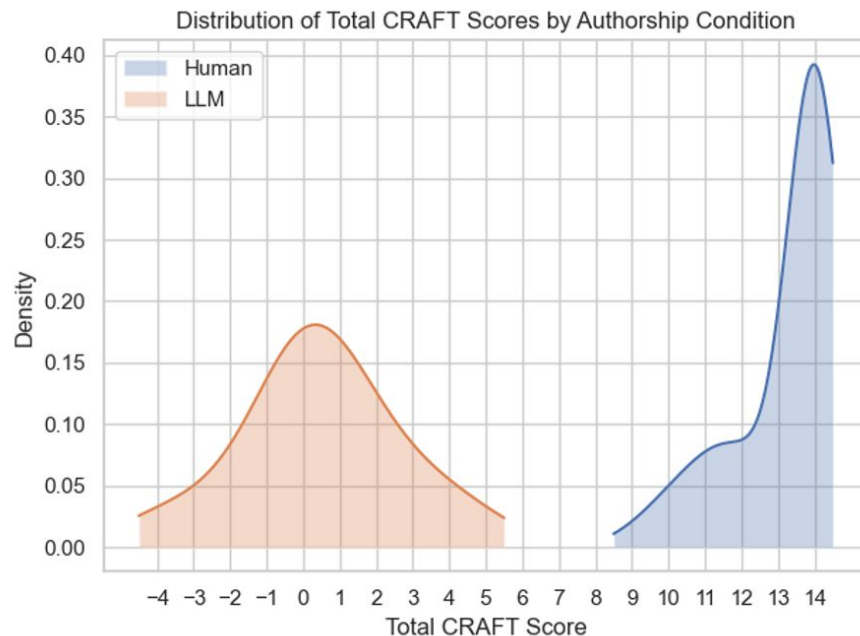
The Results

No overlap

Table 1
CRAFT Performance by Condition

Measure	Human M (SD)	LLM-generated M (SD)	ΔM	Cohen's d	95% CI	N
Structural Restoration	3.97 (0.26)	-0.32 (1.32)	4.28	3.16	[3.93, 4.63]	60
Lexical Restoration	9.28 (1.33)	0.78 (1.30)	8.50	4.65	[8.03, 8.97]	60
Total	13.25 (1.36)	0.47 (2.09)	12.78	5.34	[12.16, 13.40]	60

Note. Values are means with standard deviations in parentheses. ΔM = mean paired difference (Human - LLM-generated). Cohen's d reflects paired effect size. CI = confidence interval.



Discussion & Future Directions

Writing increasingly blends human & machine contributions; research should **map this space** using process-based measures

- Clear separation between human and LLM baselines may support measurement of future work, including hybrid texts
- Lexical and structural measures capture complementary aspects of writing, probing processes rather than surface features
- Limitations: effort effects and generalizability
- Future work: binary detection → graded authorship (incl. hybrid text)
- Practical use: baseline profiles support evidence-based, non-adversarial assessment, and writing process dialogue

Authorship is cognitive not textual, as writing encodes retrievable lexical and structural decisions that can be directly assessed.

Theoretical Contribution

Questions? Please ask!



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GitHub



References

- Aho, A., Simmons, F., Willis, C., & Pawling, R. (2020). Undergraduate students' ability to revise text effectively: Relationships with topic knowledge and working memory. *Journal of Research in Writing*, 33(1), 54–76. <https://doi.org/10.1111/1467-8817.2009.01432.x>
- Al-Abbas, T. D., Abbas, T., & Abbas, A. R. (2021). ChatGPT generated text detection based on deep learning (sequential and transform) techniques. *International Journal of Intelligent Engineering & Systems*, 18(1), 279–296.
- Altshuler, B., Jaber, R., Abu-Alrub, A., & Badar, K. (2021). Deep learning detection models for large language models: An empirical study. *Computing and Artificial Intelligence*, 33(1), 91–104. <https://doi.org/10.1007/s00521-024-10538-y>
- Al-Shabani, M. S., & Ahmed, M. (2026). Arabic machine-generated text detection: Stylistic analysis and cross-model evaluation. *Expert Systems with Applications*, 305, 130644. <https://doi.org/10.1016/j.eswa.2025.130644>
- Barkoui, K. (2019). What Can L2 Writers' Pausing Behavior Tell Us About Their L2 Writing Processes? *Studies in Second Language Acquisition*, 41(3), 529–554. <https://doi.org/10.1017/S027226319000010X>
- Barot, J., & Alameddawi, M. R. (2025). Efficacy of AI-Text Detection Tools in Distinguishing Student-Produced, AI-Edited, and AI-Generated Essays. *Technology, Knowledge and Learning*. <https://doi.org/10.1007/s10758-025-09884-0>
- Bazerman, C. (2016). *What Do Successful Students Write?* The Use of Language. In C. Bazerman, S. Graham, & I. Fitzgerald (Eds.), *Handbook of Writing* (2nd ed.). The Guilford Press.
- Benton, J. S., & Kiewra, K. A., Whitfill, J. M., & Dennison, R. (1993). Encoding and external-storage effects on writing processes. *Journal of Educational Psychology*, 85(2), 267. <https://doi.org/10.1037/0022-0663.85.2.267>
- Boud, D., & Falchikov, N. (2006). Aligning assessment with long-term learning. *Assessment & Evaluation in Higher Education*, 31(4), 399–413. <https://doi.org/10.1080/0269030060079050>
- Cheppell, C. A. (2021). Argument-based validation in testing and assessment. *Scale Publications*.
- Crook, F. I. M., & Lockhart, R. S. (1972). Levels of processing: A framework for memory research. *Journal of Verbal Learning and Verbal Behavior*, 11(6), 671–684. [https://doi.org/10.1016/S0022-5371\(72\)80001-X](https://doi.org/10.1016/S0022-5371(72)80001-X)
- Creswell, J. W., & Creswell, J. D. (2023). *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches* (6th ed.). SAGE Publications.
- Danthuri, S. S., See, A., Ghoshal, S., Huang, P.-S., McAdam, R., Welil, J., Bachani, V., Kakasali, A., Stanford, R., Matejovicova, T., Hayes, J., Vyasa, N., Merrey, M. A., Brown-Cohen, J., Bunel, R., Balke, B., Cemgil, T., Ahmed, Z., Stocopolcu, K., ... Kohl, P. (2024). Scalable watermarking for identifying large language model outputs. *Nature*, 634(8035), 818–823. <https://doi.org/10.1038/s41586-024-08025-4>
- Deane, P., O'Sullivan, N., Quinlan, T., Fowler, M., Welsh, C., & Cognigni, M. (2023). Cognitive models of writing: Using generative AI to generate synthetic text. *Journal of Experimental Psychology: Applied*, 29(1), 1–17. <https://doi.org/10.1037/xap0000409>
- Deng, R., Jiang, M., Xu, Y., Li, Y., & Liu, X. (2025). Does ChatGPT enhance student learning? A systematic review and meta-analysis of experimental studies. *Computers & Education*, 227, 105224. <https://doi.org/10.1016/j.compedu.2024.105224>
- Drexler, F., Werner, A., Lehmann, F., Hoppe, M., Schmidt, A., Buschek, D., & Welsh, R. (2024). The AI Ghostwriter Effect: When Users do Not Perceive Ownership of AI-Generated Text by Self-Declared as Authors. *ACM Transactions on Computer-Human Interaction*, 31(2), 1–40. <https://doi.org/10.1145/3658785>
- Eid, H., Yildiz, H. C., Niko, B., Ören, N. C., & Gundogdu, B. (2025). Evaluating the Efficacy of Perplexity Scores in Distinguishing AI-Generated and Human-Written Abstracts. *Academic Radiology*, 32(4), 1785–1790. <https://doi.org/10.1016/j.acr.2025.01.011>
- Ekhart, A. M., Elaid, K., & Ahmed, S. (2023). Evaluating the efficacy of AI-generated text detection tools. *Journal of Intelligent Information Systems*, 17, 105–120. <https://doi.org/10.1007/s40201-023-00140-5>
- Fernandes, D., Villa, S., Alcázar, S., Haavisto, O., Buschek, D., Schmidt, A., Kosch, T., Shen, C., & Welsh, R. (2026). It makes you smarter but none the wiser: The disconnect between performance and metacognition. *Computers in Human Behavior*, 175, 108779. <https://doi.org/10.1016/j.chbh.2025.108779>
- Fleckenstein, J., Meyer, J., Jansen, T., Keller, S. D., Köller, O., & Möller, J. (2024). Do teachers spot AI? Evaluating the detectability of AI-generated texts among student essays. *Computers and Education: Artificial Intelligence*, 6, 100209. <https://doi.org/10.1016/j.caeai.2024.100209>
- Gasser, K. C., Devkota, H., & Kretschmer, A. (2025). Detecting AI-generated text: Current methods. *Journal of Educational Technology Research*, 8(3), 255–278. <https://doi.org/10.1016/j.jetr.2025.03.001>
- Gentile, A., & Hochman, G. (2025). It's Scary to Use It, It's Scary to Refuse It: The Psychological Dimensions of AI Adoption—Anxiety, Motives, and Dependency. *Systems*, 13(2), 82. <https://doi.org/10.3390/systems13020082>
- Fulcher, G. (2024). *Practical Writing Analysis* (2nd ed.). Routledge. <https://doi.org/10.4324/97810320373629>
- Gobaitho, D., & Torrance, M. (2004). Revision in the Context of Different Drafting Strategies. In L. Allot, L. Chanquoy, & P. Lagy (Eds.), *Revision Cognitive and Instructional Processes* (Vol. 13, pp. 63–85). Springer Netherlands. https://doi.org/10.1007/978-94-007-1048-1_5
- Giordano, D., & Papaleo, D. (2020). Distinguishing Reality from AI: Generative Synthetic Content. *Computers*, 14(1), 1. <https://doi.org/10.3390/computers14010001>
- Gouail, L., Matte, M. L., Mendoza, A., Alvarado, L., & Veloso, A. (2024). AI or student writing? Analyzing the situational and linguistic characteristics of undergraduate student writing and AI-generated assignments. *Journal of Second Language Writing*, 66, 101160. <https://doi.org/10.1016/j.jslw.2024.101160>
- Grisard, M. K., Khachatryan, I. A., & Grabovay, A. V. (2024). Stock More LLMs: Efficient Detection of Machine-Generated Text via Perplexity Approximation. *Doklady Mathematics*, 110(3), 5203–5211. <https://doi.org/10.1134/S10645646240602075>
- Guo, W., Liang, Z., Wang, C., Li, X., Hu, H., Chen, S., Yu, Q., & Zhao, Q. (2025). Short-and AI collaborative creative Problem-Solving: The role of human expertise. *Computers & Education*, 239, 105433. <https://doi.org/10.1016/j.compedu.2025.105433>
- Hayes, J. R. (1996). A new framework for understanding cognition and affect in writing: Theories, methods, individual differences, and applications. (pp. 1–27). Lawrence Erlbaum Associates, Inc.
- Hayes, J. R., & Flower, L. S. (1980). Identifying the Organization of Writing Processes. In L. W. Gregg & E. R. Steinberg (Eds.), *Cognitive Processes in Writing* (pp. 3–30). Lawrence Erlbaum Associates.
- Hyland, K. (2019). *Sentence Lengthening*. Cambridge University Press.
- Janovic, J., Pekovic, S., Djokovic, R., & Vukovic, D. (2024). Assessing the Effectiveness of Academic Integrity Institutional Policies: How Can Honor Code and Severe Punishments Deter Students' Cheating—Moderating Approach? *Sage Open*, 14(4), 21582440241307430. <https://doi.org/10.1177/21582440241307430>
- Jin, Y., Yan, L., Echeverri, V., Golévic, D., & Martinez-Maldonado, R. (2025). Generative AI in higher education: A global perspective of institutional adoption policies and guidelines. *Computers and Education: Artificial Intelligence*, 8, 100548. <https://doi.org/10.1016/j.caeai.2024.100548>

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